# Railway

## Documentation Infrastructure Audit Report

Prepared by VeriOps

2026-06-26 16:06 UTC

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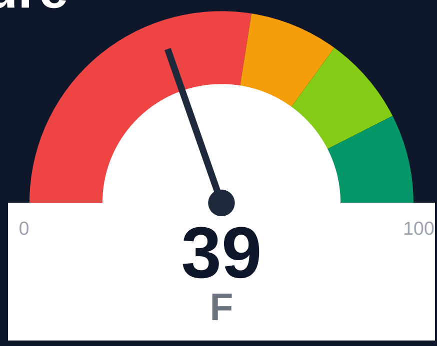



**Risk: Critical**

### Key Metrics

Pages scanned: 315

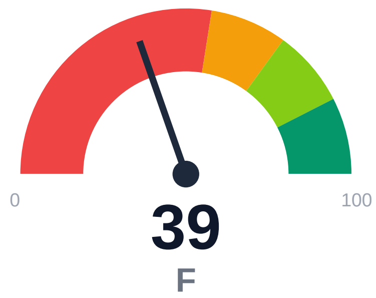
Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

### Headline Findings

- API usage-doc coverage is low: 1.95% (23 API pages detected)
- Example reliability estimate is low: 0.0%

## Executive Summary




Railway documentation audit covered 315 pages at 100% crawl coverage with zero broken links and strong retrieval readiness (91.01 overall score). Critical gaps: API reference coverage is 1.95% (only 23 API pages detected), example reliability is 0.0%, and only 30.15% of 1,078 key claims include supporting evidence. Freshness tracking covers 99.05% of pages, and 92.38% of content is chunkable for AI retrieval. Immediate value lies in expanding API documentation and validating code examples to reduce support load and improve developer onboarding.

External posture: SEO/GEO issue rate **1.9%**, public API coverage **1.9%**, broken links **0**.

| | | |
|---|---|---|
| **39** Audit Score | **F** Grade | **Critical** Risk Band |
| **315** Pages Scanned | **0** Broken Links | **1.9%** SEO Issue Rate |

### Per-Site Breakdown

| Site URL | Pages | Broken Links | API Cov % | Example Rel % | SEO Issue % |
|---|---|---|---|---|---|
| https://docs.railway.com/ | 315 | 0 | 1.9 | 0.0 | 1.9 |

## Industry Benchmark Comparison

| Benchmark | Score | Gap |
|---|---|---|
| Top performers (Stripe, Twilio) | **92+** | **-53** |
| Industry average (developer docs) | **85** | **-46** |
| Minimum acceptable | **70** | **-31** |
| **Your score** | **39** | **-** |

Gap to industry average: **46 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

## Board-Level Metrics

API Coverage (Public)

1.9%

SEO/GEO Issue Rate (lower is better)

1.9%

Example Reliability (estimate)

0.0%

Freshness Visibility (last-updated metadata)

99.0%

**Internal Metrics (from scorecard)**

API Coverage (Internal)

1.9%

Example Reliability (Internal)

0.0%

Docs-Contract Drift

69.8%

## Pipeline Opportunity Fit (Deterministic)

This section maps cold-audit signals to problem classes that the pipeline can remediate directly. No LLM inference is used in this mapping.

LLM Retrieval Readiness (deterministic proxy)

91.0%

Audit Confidence (crawl scope + verification quality)

90.0%

Readiness band: **Strong**. Confidence: **High**. Open remediation opportunities: **2**.

SEO/GEO Quality Signal (inverse issue rate)

92.4%

Example Reliability Signal

0.0%

Freshness Visibility Signal

99.0%

API Coverage Signal

1.9%

### LLM retrieval readiness breakdown (deterministic)

Chunkability

92.4%

Answerability

88.9%

Citationability

100.0%

Stale Risk (inverse)

78.9%

Evidence coverage: **30.1%** (325/1078 key claims backed by deterministic evidence signals).

### Highest-value remediation targets

| Problem | Severity | Pipeline solution path |
|---|---|---|
| Code example reliability risk | HIGH | Snippet lint + smoke checks + protocol self-verify coverage |
| API reference coverage and drift risk | HIGH | API-first contract flow, validators, regression gates, and drift checks |

## Financial Exposure Model

| Low Estimate | Base Estimate | High Estimate |
|---|---|---|
| **$171,458** annual exposure | **$265,328** annual exposure | **$422,695** annual exposure |
| Remediation: **$1,805** | Remediation: **$4,702** | Remediation: **$7,600** |
| Monthly loss: **$3,562** | Monthly loss: **$11,144** | Monthly loss: **$24,016** |
| Opportunity: **$10,575**/mo | Opportunity: **$10,575**/mo | Opportunity: **$10,575**/mo |

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

## Risk Matrix

| ID | Issue | Severity | Monthly Loss | Fix Cost |
|---|---|---|---|---|
| F-API-COVERAGE | Undocumented API operations | HIGH | $2,508 | $950 |
| F-EXAMPLES-RELIABILITY | Code examples fail or are non-runnable | HIGH | $1,938 | $808 |
| F-DRIFT | Code/docs contract drift | HIGH | $1,642 | $570 |
| F-RETRIEVAL | RAG retrieval quality risk | HIGH | $1,949 | $855 |
| F-FRESHNESS | Documentation freshness debt | MEDIUM | $969 | $475 |
| F-LAYERS | Missing required doc layers | LOW | $1,539 | $712 |
| F-TERMINOLOGY | Terminology inconsistency | LOW | $598 | $332 |

## Priority Action Plan (Next 14 Days)

| ID | Finding | Severity |
|---|---|---|
| F-API-COVERAGE | Undocumented API operations | HIGH |
| F-EXAMPLES-RELIABILITY | Code examples fail or are non-runnable | HIGH |
| F-DRIFT | Code/docs contract drift | HIGH |
| F-RETRIEVAL | RAG retrieval quality risk | HIGH |
| F-FRESHNESS | Documentation freshness debt | MEDIUM |

### Recommended Actions

**1.** Implement automated example validation for all code samples to achieve >80% reliability from current 0.0% [impact: critical, effort: medium]

**2.** Expand API reference documentation from 1.95% coverage (23 pages) to cover core REST/GraphQL endpoints and CLI commands [impact: critical, effort: high]

**3.** Add evidence links or screenshots to the 753 unsupported claims (currently 30.15% coverage) starting with onboarding and pricing pages [impact: high, effort: medium]

**4.** Audit and refresh the 21.08% of pages flagged with stale risk to align with current product state [impact: high, effort: medium]

**5.** Resolve SEO/geo issues affecting 1.9% of pages to improve regional discoverability [impact: medium, effort: low]

## Expert Analysis: Strengths and Risks

### Strengths

+ Zero broken links across all 315 pages (docs, internal, and repository navigation)
+ Strong retrieval readiness score of 91.01 with 100% citationability and 88.89% answerability for AI-assisted support
+ Near-complete freshness tracking at 99.05% last-updated coverage
+ High actionability at 90.0% coverage, indicating clear next-steps for users
+ Complete crawl coverage at 100% with no trap URLs or scope gaps

### Risks

- API reference coverage at 1.95% (23 pages) creates significant onboarding friction for developers integrating Railway programmatically
- Example reliability at 0.0% means code samples are unvalidated, likely causing copy-paste failures and support tickets
- Only 30.15% of 1,078 claims have supporting evidence, undermining trust and creating verification burden for users
- Stale risk at 21.08% indicates over one-fifth of content may be outdated despite freshness metadata
- Low API coverage may indicate detection limitations or genuine absence of REST/GraphQL/CLI reference material
- Missing evidence for 753 claims forces users to test assertions independently, slowing adoption
- SEO/geo issue rate at 1.9% may affect discoverability in regional search results

### Limitations

* External audit cannot verify functional correctness of code examples or API endpoints, only structural presence and markup patterns.
* Evidence coverage detection identifies claims but cannot assess quality, accuracy, or relevance of linked supporting materials.
* API coverage measurement depends on page structure heuristics; actual API surface may be documented in non-standard formats not detected.
* Stale risk flags are probabilistic based on metadata and content patterns, not guaranteed alignment with current product behavior.
* Crawl covers public documentation only; private or authenticated content, in-app help, and support knowledge bases are excluded.

## Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

### 1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

### 2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

### 3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

### 4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

### 5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

## Appendix A -- Economic Model Assumptions

| Assumption | Value |
|---|---|
| engineer_hourly_usd | 95.0 |
| support_hourly_usd | 55.0 |
| release_count_per_month | 8.0 |
| baseline_manual_sync_hours_per_week | 4.0 |
| avg_release_delay_hours | 6.0 |
| monthly_support_tickets | 450.0 |
| docs_related_ticket_share | 0.22 |
| avg_ticket_handling_minutes | 18.0 |
| monthly_doc_influenced_evals | 180.0 |
| eval_to_customer_rate | 0.12 |
| avg_customer_monthly_value_usd | 420.0 |
| docs_friction_revenue_sensitivity | 0.35 |

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

## Appendix B -- Evidence Traceability

| Finding | Evidence Source | Confidence |
|---|---|---|
| F-API-COVERAGE -- Undocumented API operations | OpenAPI + docs text scan | HIGH |
| F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable | examples_smoke_report.json | HIGH |
| F-DRIFT -- Code/docs contract drift | pr_docs_contract.json + api_sdk_drift_report.json | HIGH |
| F-RETRIEVAL -- RAG retrieval quality risk | retrieval_evals_report.json | HIGH |
| F-FRESHNESS -- Documentation freshness debt | frontmatter date scan | HIGH |
| F-LAYERS -- Missing required doc layers | policy pack + frontmatter layer scan | HIGH |

| | | |
|---|---|---|
| F-TERMINOLOGY -- Terminology inconsistency | glossary.yml forbidden terms scan | LOW |

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.